



Backend API Documentation

Ambulance Pre-Alert System — Node.js + Express + MySQL 8 + Socket.IO

1. Overview

GoldenHour connects ambulance crews with emergency departments. A crew creates an emergency pre-alert with priority, destination, ETA, location, chief complaint and clinical information. The hospital receives a structured live case, acknowledges it, prepares the team, tracks ambulance progress and records arrival and closure.

2. Base URL

```
http://localhost:5000
```

For live deployment, replace localhost with your server IP or domain.

3. Authentication

All protected endpoints require a JWT Bearer token in the Authorization header:

```
Authorization: Bearer <token>
```

Obtain the token by calling POST /api/v1/auth/login.

4. API Endpoints

Method	Endpoint	Description	Auth
POST	/api/v1/auth/register	Register user	No
POST	/api/v1/auth/login	Login and get token	No
GET	/api/v1/case-types	Get case type list	No
POST	/api/v1/requests	Broadcast pre-alert	No
GET	/api/v1/requests/:id	Poll request status	No
POST	/api/v1/cases	Create case	Yes
GET	/api/v1/cases/:id	Get case details	Yes
POST	/api/v1/cases/:id/send	Send pre-alert	Yes
POST	/api/v1/cases/:id/acknowledge	Hospital acknowledge	Yes
POST	/api/v1/cases/:id/location	Update ETA/location	Yes
POST	/api/v1/cases/:id/team-readiness	Record team readiness	Yes
POST	/api/v1/cases/:id/divert	Divert case	Yes
POST	/api/v1/cases/:id/arrive	Record arrival	Yes
POST	/api/v1/cases/:id/close	Close case	Yes
GET	/api/v1/cases/:id/events	Get activity timeline	Yes
POST	/api/v1/cases/:id/resources	Request resources	Yes
GET	/api/v1/hospitals/:id/incoming	Incoming cases board	Yes
PUT	/api/v1/hospitals/:id/capacity	Update hospital capacity	Yes

5. Case Lifecycle

DRAFT → SENT → DELIVERED → ACKNOWLEDGED → ARRIVED → CLOSED

Exception states: QUEUED, DIVERTED, ACK_FAILURE, CANCELLED

6. Sample Requests & Responses

6.1 Broadcast Pre-Alert

POST /api/v1/requests

Request Body:

```
{ "case_type_id": 1, "age": 45, "gender": "M", "blood_group": "O+", "vitals": {
  "systolic_bp": 82, "diastolic_bp": 50, "heart_rate": 132, "resp_rate": 26, "spo2": 88,
  "glucose": null }, "consciousness": "Semi-Conscious", "origin": { "lat": 12.9716, "lng":
  77.5946, "accuracy_m": 18 }, "broadcast_radius_km": 15, "images": [], "eta_minutes": 12,
  "notes": "Patient entrapped, one unit O- given", "ambulance_id": "KA01AB1234" }
```

Response:

```
{ "id": "GH-2026-0001", "hospitals_notified": 4, "status": "PENDING" }
```

6.2 Poll Request Status

GET /api/v1/requests/GH-2026-0001

Response (Accepted):

```
{ "id": "GH-2026-0001", "status": "ACCEPTED", "accepted_by": "City Emergency Hospital",
  "hospitals_notified": 4, "accepted_hospital": { "name": "City Emergency Hospital",
  "distance_km": 2.1, "eta_min": 8, "phone": "080-12345678", "lat": 12.9716, "lng": 77.5946
  } }
```

6.3 Acknowledge Case

POST /api/v1/cases/1/acknowledge

```
{ "notes": "ED team ready, trauma bay prepared" }
```

Response:

```
{ "success": true, "message": "Emergency alert acknowledged", "data": { "case_code":
  "GH-2026-0001", "status": "ACKNOWLEDGED", "acknowledged_by": "Emergency Desk", "facility":
  "City Emergency Hospital" } }
```

7. Priority Computation

The backend auto-computes RED/AMBER/GREEN priority from vitals. The ambulance app never sends priority.

Priority	Condition
RED (CRITICAL)	Unconscious OR systolic BP < 90 OR SpO2 < 90 OR HR > 150
AMBER (HIGH)	Semi-Conscious OR systolic BP < 110 OR SpO2 < 95
GREEN (MEDIUM)	Conscious with normal vitals

8. Hospital Matching

The backend uses the Haversine formula to find all hospitals within broadcast_radius_km of the ambulance origin coordinates. All nearby hospitals are notified simultaneously. The first hospital to acknowledge wins — this is handled atomically.

9. Socket.IO Real-Time Events

Event	Direction	Description
join_case	Client → Server	Ambulance joins case room
join_hospital	Client → Server	Hospital joins hospital room
location_update	Client → Server	Ambulance sends live location
ambulance_location	Server → Hospital	Live ambulance position
case_acknowledged	Server → Ambulance	Hospital accepted the case
incoming_case	Server → Hospital	New case alert on board

10. Database Schema

MySQL 8 — 13 Tables:

Table	Purpose
users	Crew and hospital staff accounts
hospitals	Registered hospitals with coordinates
ambulances	Registered ambulances
cases	Core case lifecycle
case_clinical_data	ATMIST clinical handover data
critical_flags	Shock, hypoxia, cardiac arrest flags
location_updates	Live ambulance location history
acknowledgements	Hospital acceptance records
team_readiness	Trauma team preparation
hospital_capacity	Resus bays, CT, OT, blood, ventilators
resource_requests	Blood, imaging, team requests
activity_events	Immutable audit log
notifications	In-app/SMS/email notifications

11. Security

JWT authentication with 24-hour token expiry. Passwords hashed with bcrypt (salt rounds: 10). Role-based authorization (AMBULANCE_CREW, HOSPITAL_STAFF, ADMIN). Parameterized SQL queries to prevent injection. CORS enabled. Privacy-by-design: age band and sex only, no full patient identity.

12. Technology Stack

Layer	Technology
Runtime	Node.js v24
Framework	Express.js
Database	MySQL 8
Real-time	Socket.IO
Authentication	JWT + bcrypt
Mobile Frontend	React Native + Capacitor

